Angela Zhu

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Research Interests — Computer Vision and Machine Learning; Computational Sustainability; Public Interest Technology

Education

Oregon State University, ***PhD. Artificial Intelligence*** **2025 - present**
Advisor: Prof. Rebecca Hutchinson

University of Massachusetts Amherst, ***M.S. Computer Science*** **2023 - 2025**
Research: Species distribution models built from citizen science data

University of Illinois at Urbana-Champaign, ***B.S. Mathematics*** **2014 - 2018**

Research Experience

Oregon State University **2025-present**
Graduate Research Assistant *Corvallis, OR*

- Researching species distribution modeling in the MLQuESt Lab under Prof. Rebecca Hutchinson, bridging machine learning and ecological modeling approaches.
- Investigating occupancy modeling frameworks that account for imperfect detection in observation data.
- Coursework in probabilistic graphical models and deep learning.

University of Massachusetts Amherst **2024 – 2025**
Research Assistant with Prof. Subhransu Maji and Prof. Grant Van Horn *Amherst, MA*

Investigating Different Geo Priors for Image Classification

- Evaluated location-based species distribution model SINR as geographical priors for image classification, helping distinguish visually similar species with geographically distinct ranges.
- Found that combining two models and handling of unseen species can lead to significant unintended consequences in performance.
- First author non-archival paper presented at the 12th Fine-Grained Visual Categorization (FGVC) Workshop at CVPR 2025.

Few Shot Range Estimation

- Investigated the impact of habitat and range text on model outputs, identifying critical factors influencing prediction accuracy, incorporated into ablations of paper.
- Co-author on resulting paper; contributed to paper by creating visualizations of species ranges over varying context points and text inputs.

Center for Data Science at UMass Amherst **May 2024 – Aug 2024**
Research Intern - Data Science for the Common Good Fellow *Amherst, MA*

GeoModel and iNaturalist

- Built a human in the loop platform for experts to collect expert species species data and improve species distribution model accuracy in collaboration with computer vision faculty and researchers and iNaturalist engineers
- Developed and deployed (Docker) a web application that enables experts to annotate species range maps using predictions from the SINR GeoModel- ML-based range prediction model to refine species distributions
- Designed interactive mapping interface (React JS, Python) and database to store predictions and annotations
- Conducted 5 successful user studies with stakeholders to get feedback on the platform and identified future research directions

Publications and Presentations

- First Author of "**Investigating Different Geo Priors for Image Classification**" non-archival paper and poster presentation at the 12th Fine-Grained Visual Categorization FGVC Workshop, CVPR, 2025.
- Coauthor "**Feedfoward Few-shot Species Range Estimation**" paper published in ICML 2025
- Poster presenter of "**iNatator: Obtaining Expert Feedback on Species Ranges**" at New England Computer Vision NEVC Workshop 2024

Industry Experience

Microsoft **2018 – 2023**
Software Engineer II *Redmond, WA*

Backend engineer for Substrate Microsoft 365

- SPARTAN team within data service organization for Office products worked on projects that the VP wanted to accelerate
- Experienced technical lead and mentor for engineering projects. Projects I worked on include:
- Building a chaos engineering platform to test the resiliency of internal services
- Built a rate limiting service for M365 data ingestion pipeline - billions of updates per day
- Defined REST APIs and written client SDKs for products.
- Shipping & deploying code to production in large ecosystem. Built, deployed, tested and packaged code
- Designed 30 day engineering onboarding curriculum for the Substrate organization
- Onboarded 15+ new hires, individually mentored 5 engineers
- Chosen to speak at internal Microsoft Aspire Conference: Team Switch Panel alongside managers to 50 employees

Expedia Group **May 2017 – Aug 2017**
Hotwire - Data Science Intern *Seattle, WA*

Added Neighborhood Ranking Feature to Hotel Search Algorithm

- Improved Hotwire's hotel sorting algorithm by adding a feature that measures popularity of neighborhoods based on search criteria. Used Hive to extract features from click data, Python for feature engineering and boosted decision trees (BDT) to train the model
- Increased overall ranking quality (NDCG) by 3%

Skills

Software Tools	Git, Visual Studio, Azure DevOps, CICD	**Cloud**	UMass Unity Cluster, AWS
Languages	Python (Familiar: R, Java, C++)	**Testing**	Pytest, MSTest, xUnit

Teaching Experience

- AI Capstone (Winter 2026)
- Introduction to Algorithms (Fall 2025)

Volunteering and Community Engagement

Heart of the Valley Running Club **2025-2026**
UMass Birdwatching Club **2023-2025**
Microsoft **2018-2023**

- Led and organized knowledge sharing and team building activities: led How-To talks, organized bookclubs and volunteering events

Grassroots Ecology Nonprofit **2020-2023**

- Habitat restoration and maintained natural ecosystems